Jay Shah

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Summary

M.S. Computer Science graduate with 3+ years of **Python development** experience across backend services, full-stack applications, and AI integrations. Proficient in **FastAPI, Flask, GraphQL, React, Next.js, Remix, Tailwind, MUI, PostgreSQL, Podman**, and **microservices on Linux**. Experienced translating client requirements into working software, writing clean well-tested code, performing unit and integration testing, and delivering thorough technical documentation.

Work Experience

University of Texas, Arlington Aug 2024 – May 2026
Graduate Assistant — Python Development and AI Systems *Arlington, TX*

- 60+ software projects had no shared development standard – client requirements weren't being consistently translated into features, code quality varied widely, and there was no unit testing or documentation process.
- Wrote modular OOP **Python** services; built **FastAPI** REST and **GraphQL** endpoints; implemented **pytest** unit and integration tests; designed **React** dashboards; authored technical handbooks and API documentation for faculty and cross-functional teams on **Linux**.
- **30% fewer failures**, **25% faster** delivery – clean, well-tested, fully documented Python code that any developer could onboard to, maintain, and extend independently.

Tata Consultancy Services Jan 2024 – May 2024
Python Developer Intern — Backend and Full Stack *Gandhinagar, India*

- Client needed new application features across **500K+ records** but requirements weren't translated into specifications – backend services had broken integrations, and there was no testing or documentation for the existing codebase.
- Understood client requirements and translated them into **FastAPI + Flask** features; designed **PostgreSQL** schemas; built **React** UI components with **Tailwind CSS**; wrote **pytest** unit and integration tests; troubleshot and debugged across the stack; deployed as **microservices** on **Linux** via **Docker**; authored technical documents and handbooks.
- **2 application features** shipped to production, system **30% more reliable**, delivery **15% faster** – fully documented, tested Python codebase delivered to client specification.

Celebal Technologies May 2023 – Sept 2023
Python Developer Intern — Full Stack *Jaipur, India*

- Three clients needed custom application features built to spec – UI prototypes didn't match requirements, existing Python code was undocumented, and there was no testing process to validate new features before deployment.
- Designed UI/UX prototypes; built **React + Tailwind CSS** frontend; developed **Flask + Python** backend services with **PostgreSQL** data layers; performed unit and integration testing; troubleshot and debugged across three client stacks; authored technical documentation and handbooks for each deployment.
- Manual workload reduced **25%**, data accuracy improved **18%** – application features delivered on time to specification across three clients, each with unique requirements.

Projects

Full-Stack Python Web Application · FastAPI · GraphQL · Next.js · React · MUI · Tailwind · PostgreSQL · Podman · Linux · pytest Dec 2025 – Mar 2026

- Client needed a new application built to specification – data was scattered, there was no API layer, and the existing UI didn't match what end users actually needed to do their jobs.
- Translated client requirements into detailed specs; designed UI/UX prototypes; built **FastAPI** backend with **GraphQL** API; developed **Next.js + React** frontend with **MUI** components and **Tailwind CSS**; used **PostgreSQL** for storage; containerized with **Podman** on **Linux**; wrote **pytest** unit and integration tests; authored full technical documentation and API handbook.
- Application processed **300K+ records** reliably, query performance **35% faster** – complete, tested, documented Python application delivered to client specification.

Python Microservices Platform · FastAPI · Flask · GraphQL · PostgreSQL · Redis · Podman · Linux · GitHub Actions · pytest Apr – Jul 2025

- A monolithic Python application handling **12K+ concurrent requests** was fragile – a single change could break unrelated features, deployment was risky, and there was no testing infrastructure to catch regressions before users hit them.
- Refactored into **Python microservices** using **FastAPI** and **Flask**; added **GraphQL** API gateway; implemented **Redis** caching; containerized with **Podman** on **Linux**; wrote comprehensive **pytest** tests; set up **GitHub Actions** CI/CD; developed technical handbooks documenting each service's design and API contracts.
- Throughput improved **40%**, **99.9% uptime** – independent service deployments let developers change and test one service without affecting the rest.

AI-Powered Python Application with Modern Frontend · FastAPI · Remix · React · Tailwind · MUI · PostgreSQL · LangChain · Podman · Linux Mar – May 2024

- **1M+ records** were inaccessible to business users – every data question required a developer, the frontend was outdated, and there was no automation layer for repetitive tasks.
- Built **FastAPI** Python backend with **LangChain** AI integration; developed **Remix + React** frontend with **MUI** and **Tailwind CSS**; designed **PostgreSQL** schema; containerized with **Podman** on **Linux**; wrote unit and integration tests; produced complete technical documentation and user handbook.
- Developer query workload reduced **40%**, application handled **1M+ requests** reliably – fully documented, tested Python application that business users could operate independently.

Education

University of Texas, Arlington — Aug 2024 – May 2026
M.S. Computer Science ***GPA: 3.90/4.00*** · *Algorithms · Distributed Systems · Software Engineering · Cloud Computing · Machine Learning* — *Arlington, TX*

Charusat University — Oct 2020 – May 2024
B.S. Computer Engineering ***GPA: 3.78/4.00*** · *Data Structures · Database Systems · OOP · Software Engineering · Computer Networks* — *India*

Skills

Python: FastAPI · Flask · GraphQL · SQLAlchemy · Celery · pytest · OOP · async/await

Databases: PostgreSQL · MySQL · Redis · MongoDB · query optimization · schema design

APIs: GraphQL · REST · FastAPI · Flask · microservices · API design

Frontend: React · Next.js · Remix · TypeScript · MUI · T

DevOps: Podman · Docker · Linux · GitHub Actions · C

AI / ML: LangChain · OpenAI APIs · PyTorch · scikit-lea